

Case Study: NHS Scotland Hospital Bed Availability for Cancer Treatment

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INTRODUCTION

The COVID19 challenges have put some strain on the operating model of NHS Scotland, which has led to ineffective operations in the healthcare sector. In order to improve how the NHS responds to patient treatments and plan their response times accordingly, a clear and strategic approach must be taken in order to successfully manage the treatment of cancer, specifically Urological Cancer.

PART ONE

SECTION A: FORECASTING EVALUATION

In every life situation, human predicts the future based on the current or past incidents that occurred in society (Hyndman and Athanasopoulos, 2021). Forecasting is the process of adopting the historical trend or data to predict particular estimations for specific managerial or business trends.

Four Key Questions that should be considered by the management at NHS

These questions include:

1. **To what extent has the forecasting technique used operational indicators and inputs?** As a consultant, the performance measure can be derived through key operational indicators and inputs. It will enable NHS Scotland to determine its performance and helps direct the forecasting technique in considering the operational indicators toward predicting the future trend for the organisation.
2. **What is the level of automation integrated with the technique?** Integrating automation to the model applied in forecasting techniques can enable NHS Scotland to maintain a better predictive process or estimate future trends (Lightner, 2018). As a consultant, this is a key criterion to consider in the forecasting technique for NHS Scotland because automation helps projection to include machine learning, intelligent technologies, and artificial intelligence.
3. **How does the technique create the variable and factors involved?** there is a need to consider the process of identifying variables and factors involved in projecting future trends in the organisation for effective management of patients, staff, and hospitals to reduce the waiting time of cancer patients in hospitals.
4. **Is this technique thoroughly effective for its purpose?** The effectiveness of the forecast technique ensures the level of accuracy achieved in the process of determining the future trend or projection in NHS Scotland. Key metrics towards

forecasting in the organisation require complete accuracy to employ a forecasting technique.

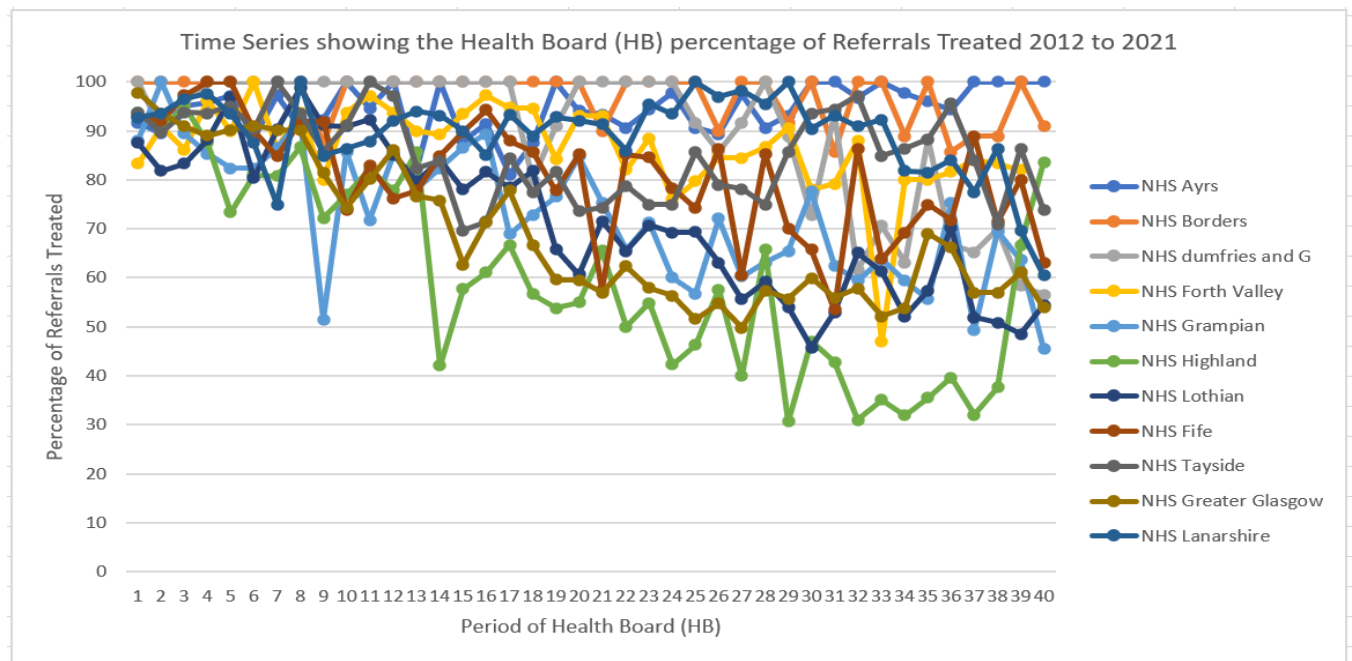
Challenges/Implications of using the Forecasting Technique

Concerning challenges or implications, organisations experience these based on the modalities involved in forecasting techniques. Some of these challenges/implications are subjected to a dynamic situation of projective trends, which include the type of data, purpose of data and so on. However, overall implications or challenges include selecting appropriate forecasting techniques, maintaining the forecasting model employed, collating the exact data type for forecasting, governmental policies and implications and economic conditions.

SECTION B: DATA ACCESS, CLEANING AND PREPARATION

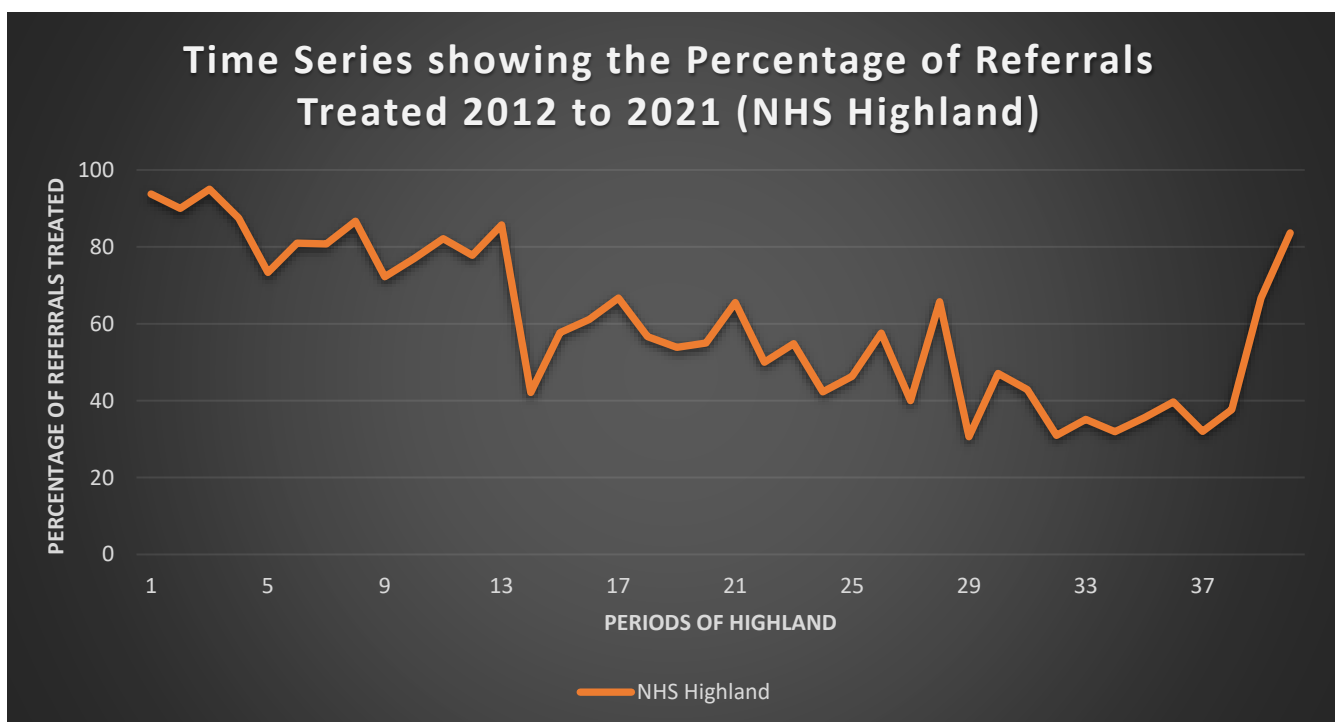
As seen in the figure below, the various locations in NHS Scotland show a different pattern of times series. This is because most cities experienced horizontal and trend & seasonal with trends patterns in the referral and treatment of patients with urological cancer. NHS Highland experienced the lowest percentage of referrals treated but experienced a huge spike in numbers from 2020-2021 while NHS Grampian, NHS Lothian experienced a decrease in referrals during 2020-2021. This implies that some health boards experienced low referrals of these patients from 2015 to 2019 before the COVID-19 period. However, there was an increase in the rate of treatment from 2020 to date, as seen in most of the health boards of NHS Scotland. According to the health update of the Public Health Service (2022), updates reveal that over 141,572 adults were diagnosed with cancer treatment which correlates with the trends shown in the above figure on the increasing treatment of cancer in the NHS Scotland. Based on this, recommendations can be made to increase the accessibility to urological cancer treatment despite the pandemic in the health sector and also develop guidelines that can accommodate more treatment of urological cancer patients in the health center.

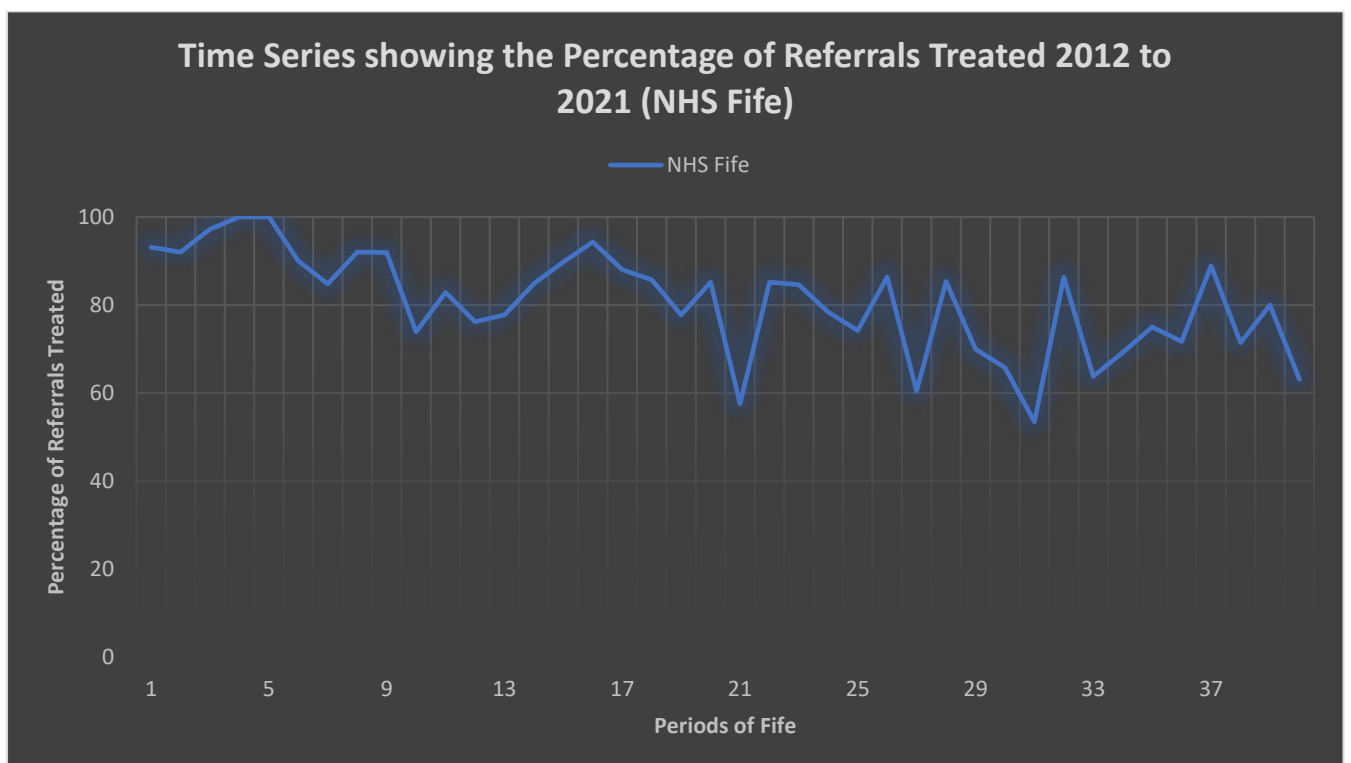
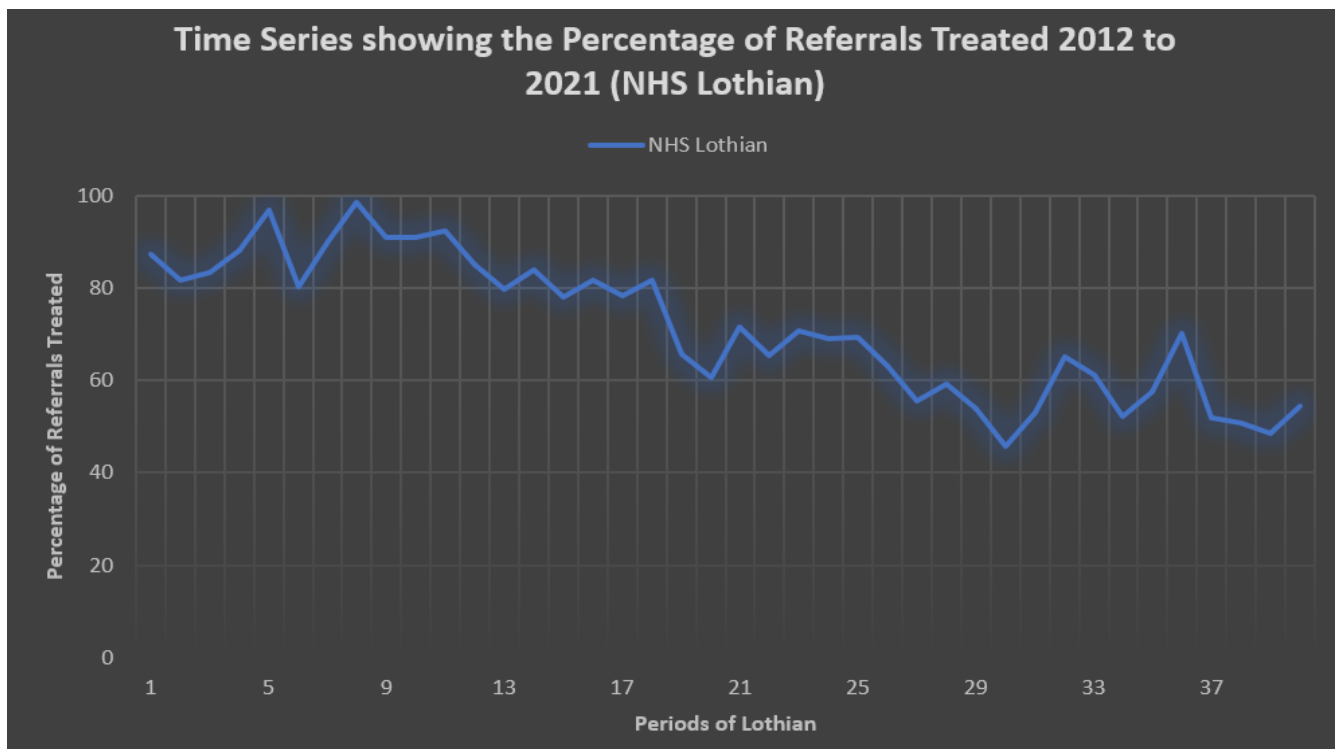
The below is the time series showing all Health Board Percentage of Referrals Treated from 2012 to 2021 in Scotland.

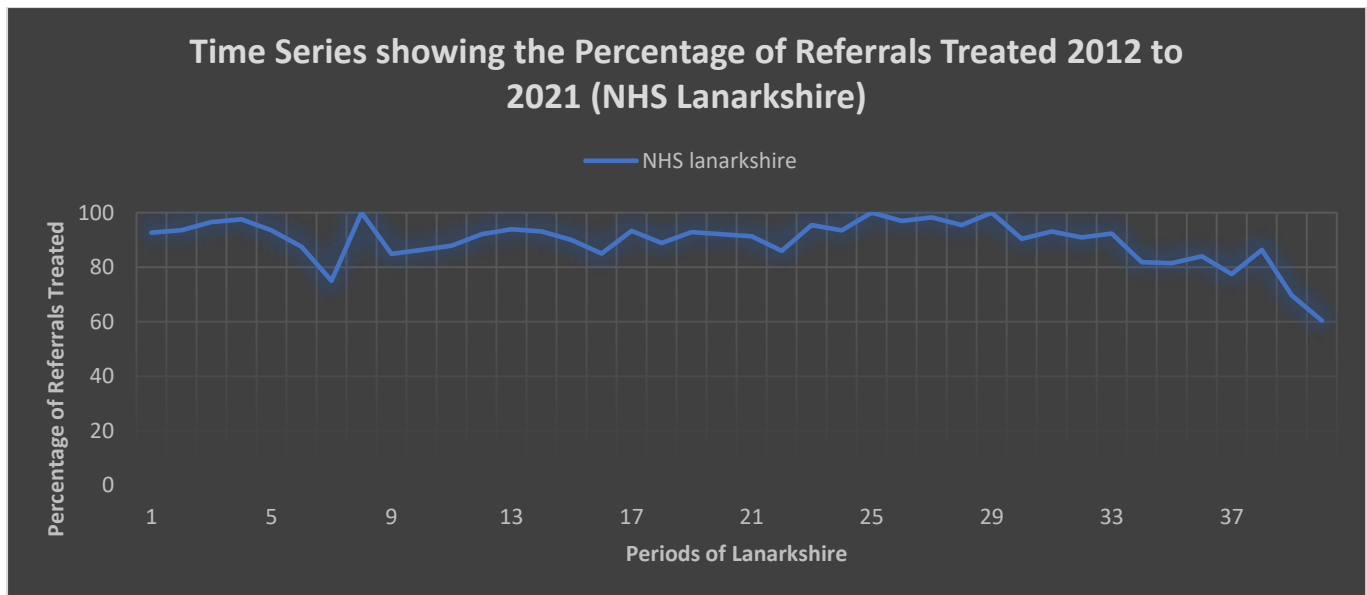


SECTION C: TIME SERIES FORECASTING

The below figures present the time series of four major health boards which NHS Highland, NHS Lothian, NHS Fife and NHS Lanarkshire that is based on forecasting analysis:





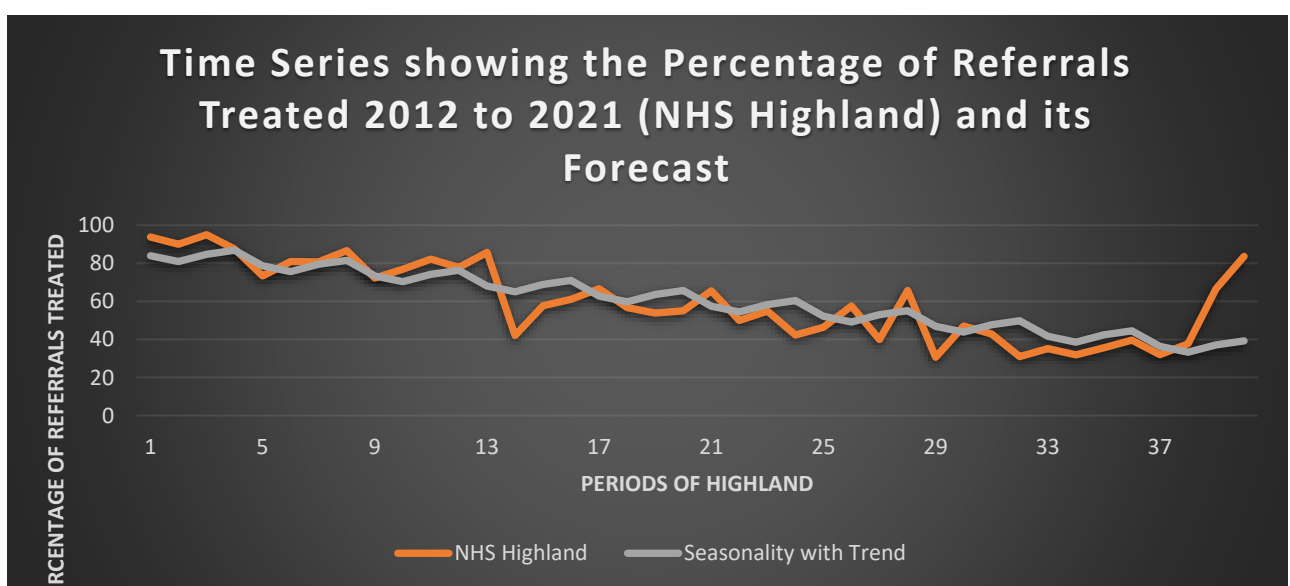


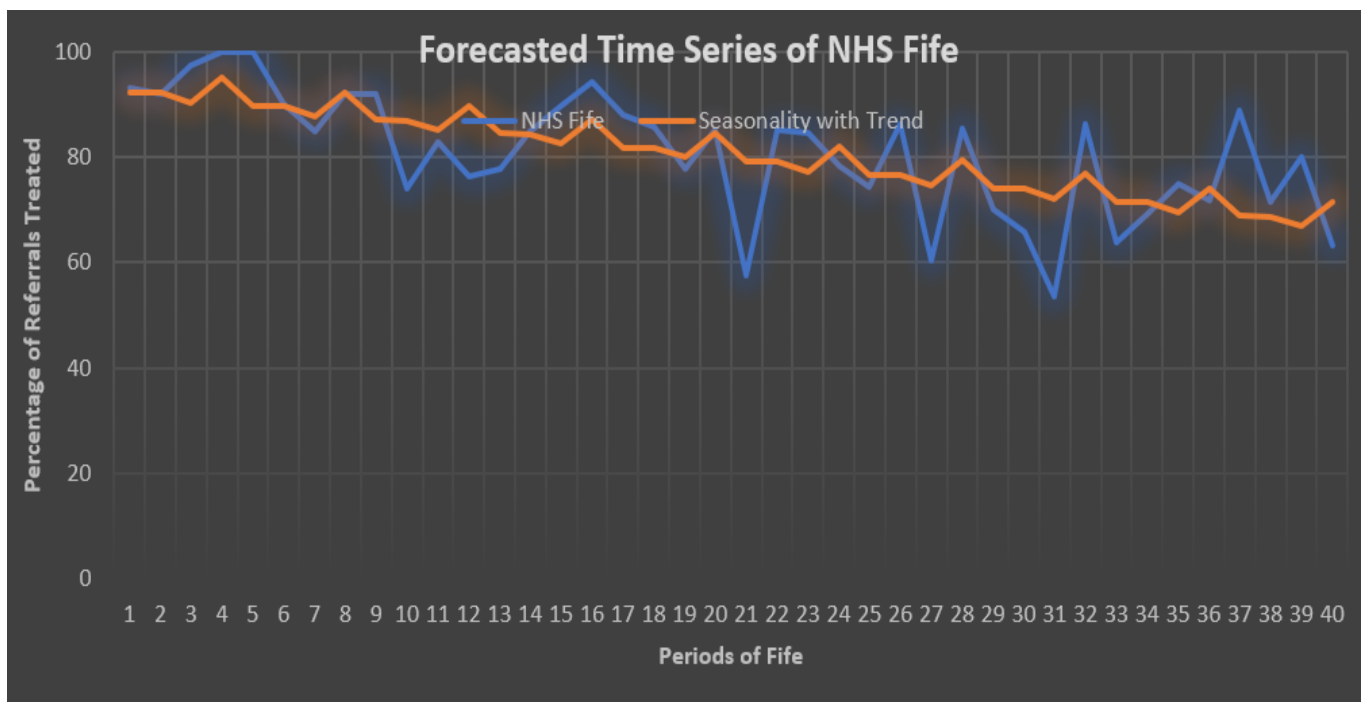
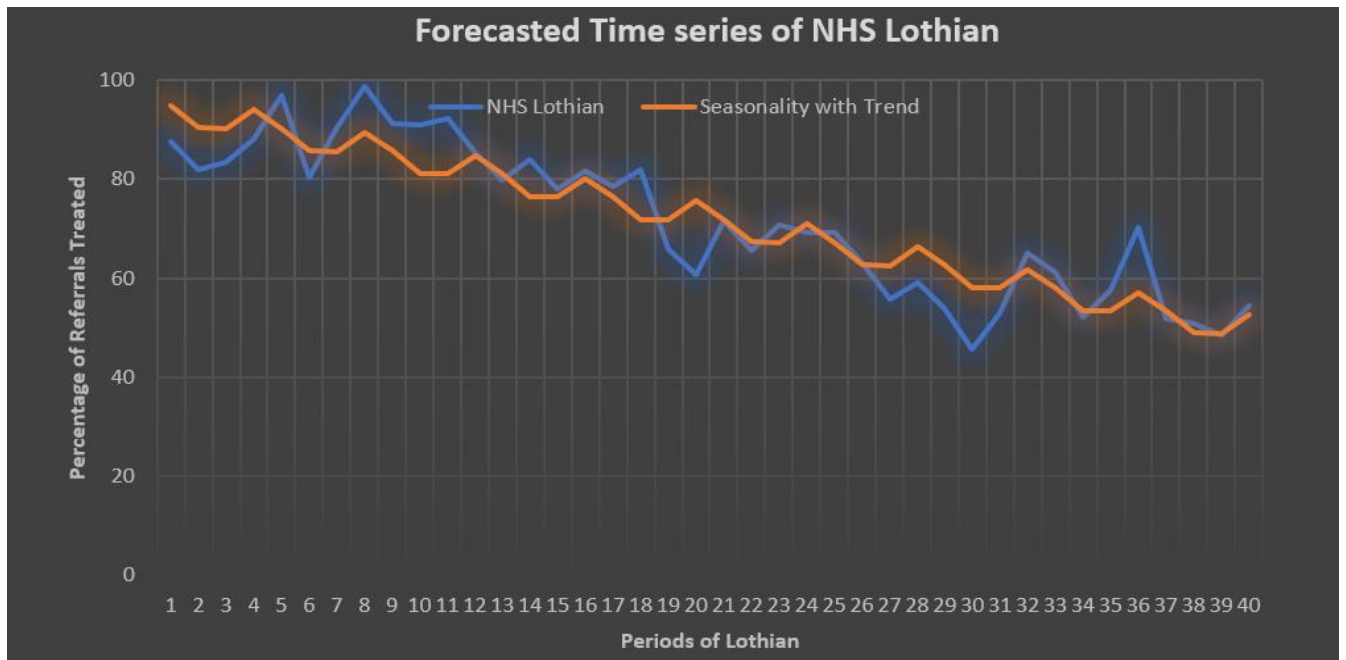
Time Series Forecasting for Each Health Boards

The following table present the pattern of the time series forecasting for the NHS Highland, Lothian, Fife and Lanarkshire in Scotland and the time series forecasting:

SN	Time Series	Patterns
1	NHS Highland	Cynical Trend Pattern
2	NHS Lothian	Seasonal trend Pattern
3	NHS Fife	Seasonal Trend Pattern
4	NHS Lanarkshire	Horizontal Pattern

Forecasted Time Series from 2012 to 2021





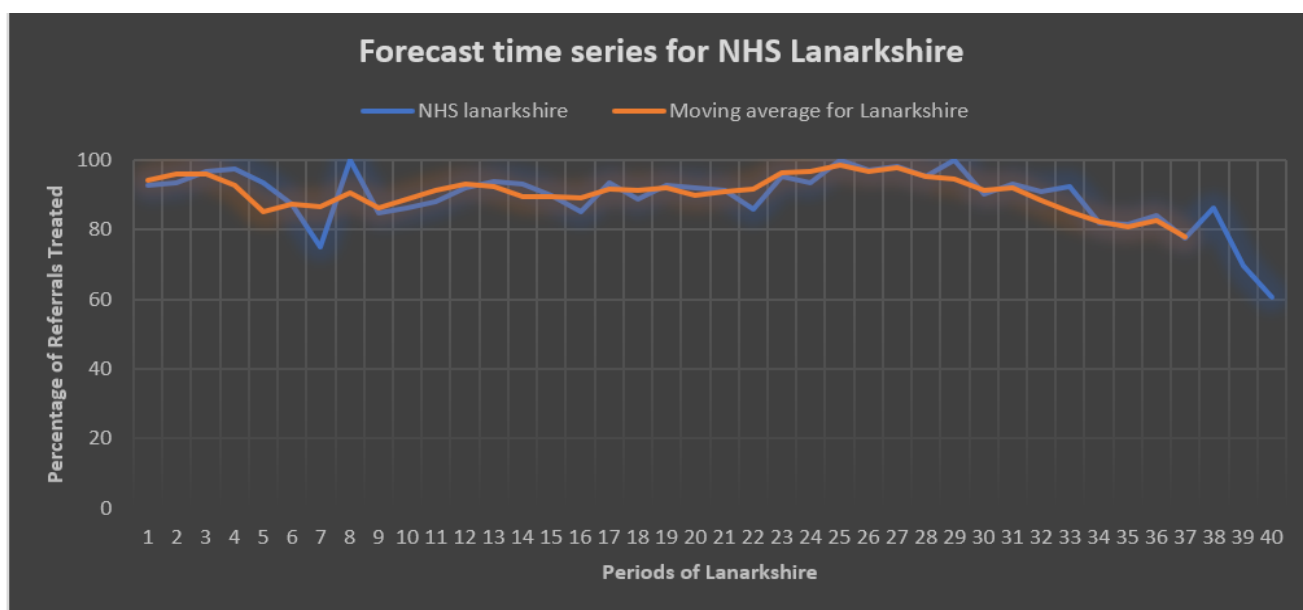


Table summary for the forecasted Time Series analysis 2012 to 2021

Health Board	Pattern Identified	Forecasting Method	Mean Squared Error
NHS Highland	Seasonal with Trend Pattern	Regression Analysis	156.86
NHS Lothian	Seasonal with Trend Pattern	Regression Analysis	41.58
NHS Fife	Seasonal with Trend Pattern	Regression Analysis	74.05
NHS Lanarkshire	Horizontal Pattern	Moving Average Method	46.22

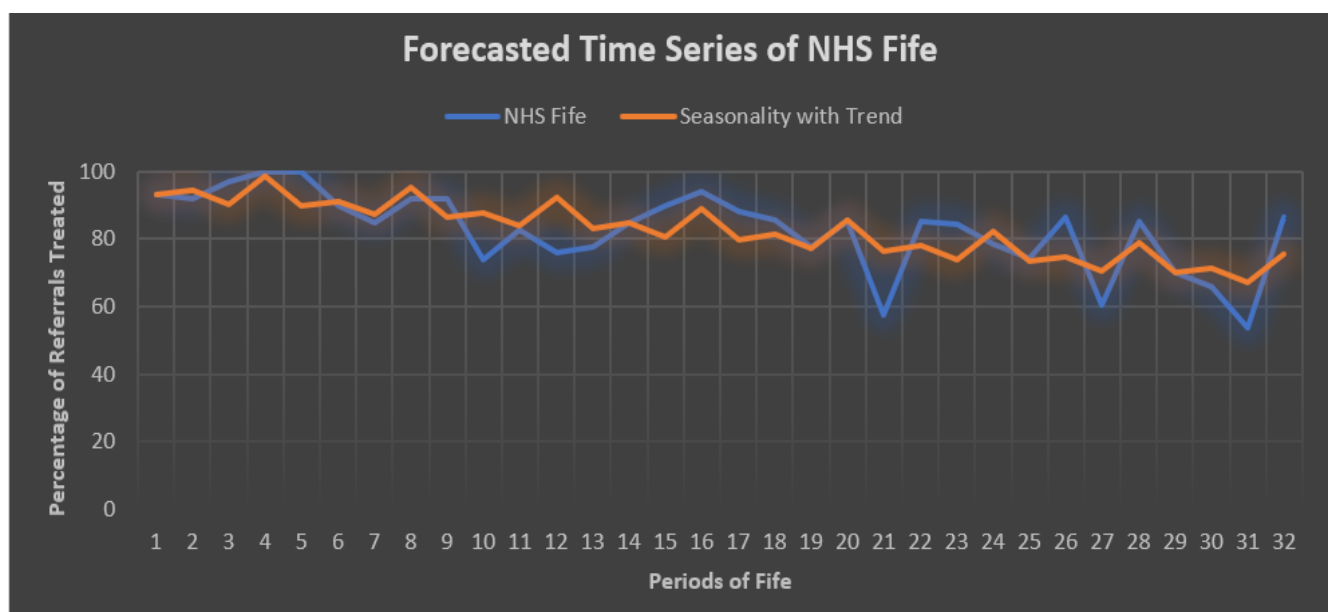
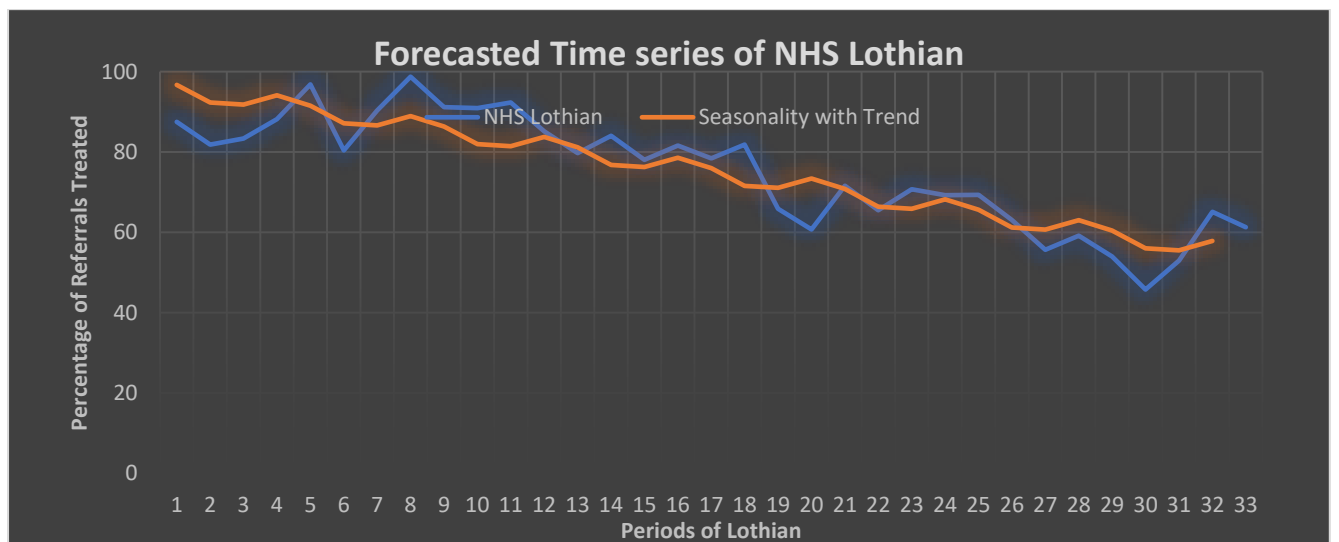
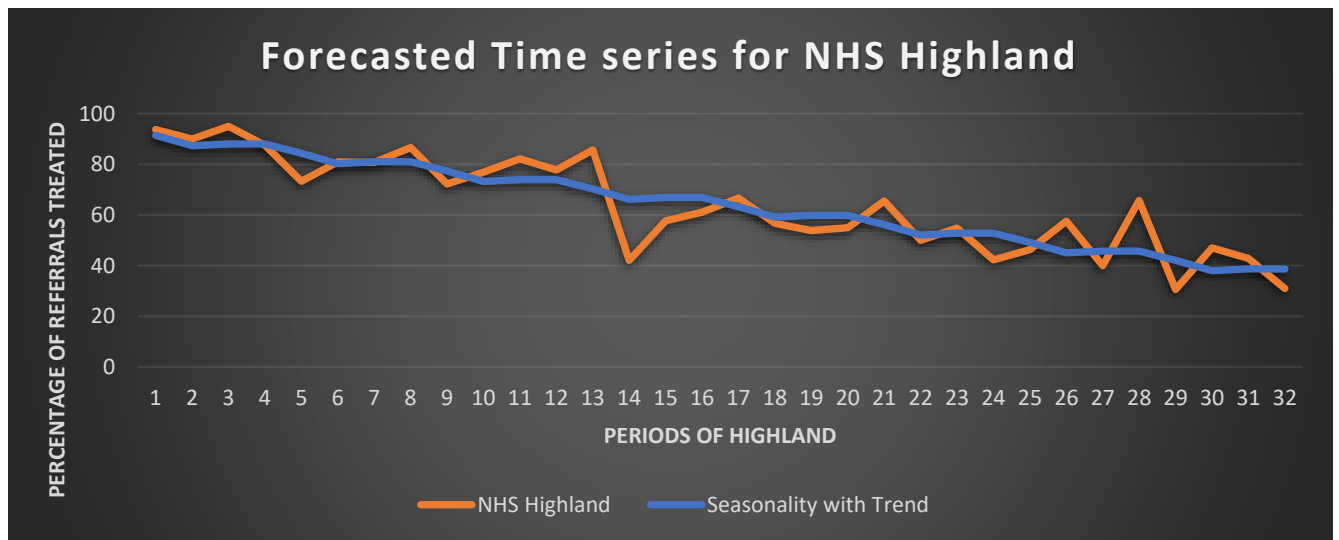
Observation in the time series forecasting analysis of data with 2020 -2021

During the analysis of this data the aim was to get the best method to predict the number of referrals coming into the NHS Scotland's system, three different methods were adopted for this analysis; the moving average method, Trend method, and Seasonality with Trend methods were used. The method which provided the lowest MSE is the one to be preferred to the management to predict the number of referrals and how resources can be allocated and managed efficiently.

NHS Highland gave 156.86 with the seasonal Trend pattern as the lowest MSE in the Regression analysis used. Likewise, NHS Lothian and NHS Fife have the same trend patterns with the lowest MSE being 41.58 and 74.05 respectively.

NHS Lanarkshire was the only health board analyzed that gave the moving average method as its lowest MSE with a value of 46.22 following a horizontal pattern.

Forecasted Time Series from 2012 to 2019 (Without 2020 and 2021)



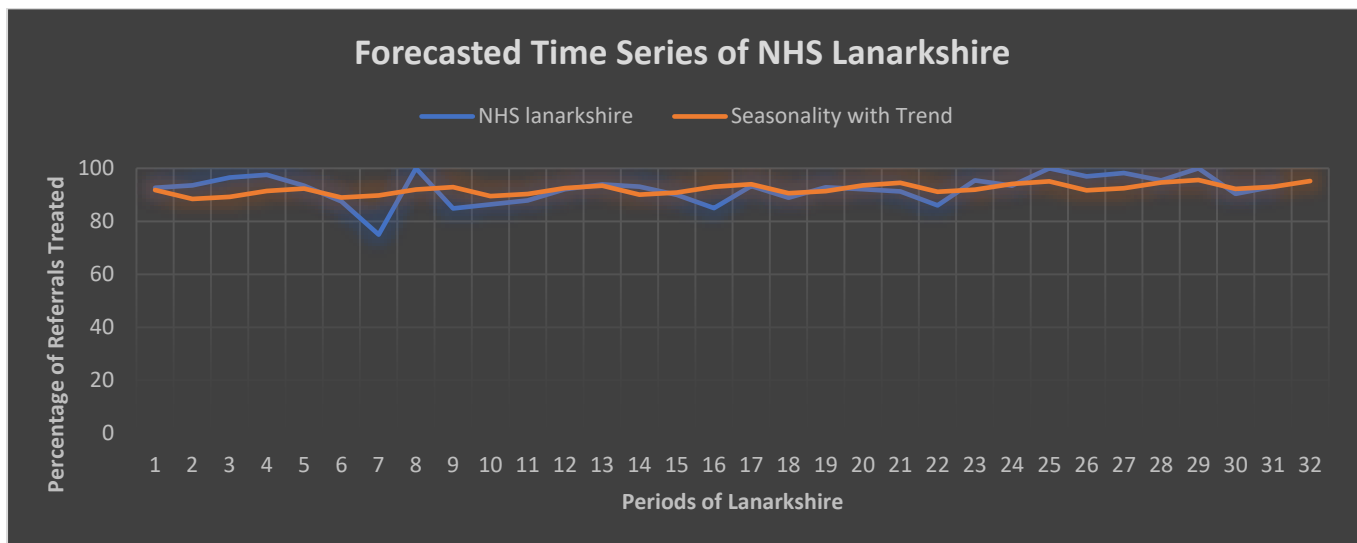


Table summary for the forecasted Time Series analysis 2012 to 2019 (Without 2020 and 2021)

Health Board	Pattern Identified	Forecasting Method	Mean Squared Error
NHS Highland	Seasonal with trend	Regression Analysis	76.39
NHS Lothian	Seasonal with trend	Regression Analysis	42.64
NHS Fife	Seasonal with trend	Regression Analysis	63.97
NHS Lanarkshire	Seasonal with trend	Regression Analysis	23.28

Observation in the time series forecasting analysis of data without 2020 -2021

The forecast analysis followed the same trend on the actual time series with little difference among the four health boards of NHS Scotland; Highland, Lothian, Fife, and Lanarkshire from 2012 to 2019. Three of the health boards had the same pattern with 2020&2021 and without, which shows little to no changes in the trends in pattern referrals for the periods. While NHS Lanarkshire was the only health board that had a pattern change while analyzing for 2020-2021, it changed the pattern from horizontal back to seasonality with Trend. To get the best MSE to analyse the forecast of the data, 3 different methods were adopted which are the Moving Average, Seasonality with trend and trend were used in calculating the forecast, this enables the analysis to adopt the best strategy in determining the forecast time series for the following years examined and the one with the lowest forecast was picked as the best MSE to predict the referrals for the NHS board.

The MSE of the health boards had lower values after analyzing compared to the data that included the years 2020-2021 except for health board NHS Lothian which had a slight increase in its MSE value of 1.06 when the data for 2020/21 was removed.

SECTION D: LINEAR PROGRAMMING/LINEAR OPTIMISATION

Optimization mathematical modeling's function in NHS Scotland's staff-available bed management

Optimisation modeling is a solution technique designed to cater to varieties of situations in the health, business, education, governance, and other sectors. In the situation of managing the available bed in NHS Scotland, modeling enables a sound decision regarding the situation of NHS Scotland. It involves a mathematical process aimed at maximizing or minimizing the complex situation in making sound and smart decisions. It aids in dispelling questions about patient attrition rates and effective retention strategies. Also, to enhance the quality and compliance of therapy while addressing scheduling issues. Major components to consider in the case of NHS Scotland include decision variables (staffs' time shift in correlation to the occupancy of bed), result variable (Optimised Time Shift among Staff) and uncontrollable variable.

Liner Programming on Problem Case on Staff Allocation in the NHS

Decision Variables

The key decision variables in the case of Tayside Hospital in Scotland include the staffs in their time shift in correlation to the occupancy of bed. Therefore lets.

S_1 = Staffs in period shift between 9:00 to 12:00

S_2 = Staffs in period shift between 12:00 to 15:00

S_3 = Staffs in time period shift between 15:00 to 18:00

S_4 = Staffs in period shift between 18:00 to 21:00

S_5 = Staffs in period shift between 21:00 to 24:00

S_6 = Staffs in period shift between 0:00 to 03:00

S_7 = Staffs in period shift between 03:00 to 06:00

S_8 = Staffs in period shift between 06:00 to 09:00

S_1 = employees who commence work at 9am & work shifts (9am - Noon), (Noon to 3pm) & (3pm to 6pm),

S_2 = employees who commence work at 12noon & work shifts (Noon to 3pm), (3pm to 6pm) & (6pm to 9pm),

S_3 = employees who commence work at 3pm & work shifts (3pm to 6pm), (6pm to 9pm) & (9pm to 12am),

S_4 = employees who commence work at 6pm & work shifts (6pm to 9pm), (9pm to 12am) & (12am to 3am),

S_5 = employees who commence work at 9pm & work shifts (9pm to 12am), (12am to 3am) & (3am to 6am),

S_6 = employees who commence work at 12am & work shifts (12am to 3am), (3am to 6am) & (6am to 9am),

S_7 = employees who commence work 3am & work shifts (3am to 6am), (6am to 9am) & (9am to Noon),

S_8 = employees who commence work at 6am & work shifts (6am to 9am), (9am to Noon) & (Noon to 3pm).

Result Variable

Results (Optimised Time Shift among Staff) = O

Objective Function

$$O = S_1 + S_2 + S_3 + S_4 + S_5 + S_6 + S_7 + S_8$$

Uncontrollable Variables

$$S_1 + S_2 + S_3 \geq 22$$

$$S_2 + S_3 + S_4 \geq 30$$

$$S_3 + S_4 + S_5 \geq 16$$

$$S_4 + S_5 + S_6 \geq 4$$

$$S_5 + S_6 + S_7 \geq 5$$

$$S_6 + S_7 + S_8 \geq 18$$

$$S_7 + S_8 + S_1 \geq 36$$

$$S_8 + S_1 + S_2 \geq 28$$

$$S_1 \geq 0,$$

$$S_2 \geq 0,$$

Non-negative constraints: $S_3 \geq 0, S_4 \geq 0, S_5 \geq 0, S_6 \geq 0, S_7 \geq 0$, and $S_8 \geq 0$

PART 2 BUSINESS INTELLIGENCE SYSTEMS - TABLEAU

Tableau Prep

The following data documents were uploaded into the tableau prep builder in order to give more information about 62 day standard cancer waiting times: names of the current hospitals, information about the health boards, geographical codes, and 62-day waiting standards for cancer waiting times recorded.

The cleaning was done in different steps with the first step of merging the special health board and hb14 dataset by a union where 2 rows not needed were removed, followed by 3 different join functions; join1(right join), join 3(left join) and join 5(middle join) were the files were merged and cleaned to removed unwanted values and dates such as null values, fields with QF and HBT were removed also, changing quarter to date format and renaming it as Time, adjusting my range of dates, filtering the cancer values to show for only urological cancer types etc. I then cleaned the data as seen in step 6 where HB and SHB fields were removed and postcode was split and cleaned. The final results of the cleaning steps can be seen in output where 8 rows remained of the cleaned data to be examined. The steps can be seen in the screenshots below.

Step One

The screenshot displays the Tableau Prep Builder interface. On the left, the 'Connections' pane lists several data sources: 'hb14_hb19.xlsx' (Microsoft Excel), 'special-health-board...' (Microsoft Excel), 'current-hospital_fla...' (Text File), 'cwt_62_day_stand...' (Text File), and 'geography_codes_a...' (Text File). Below this, the 'Tables' pane shows 'geography_codes_an...'. The main workspace shows a data flow starting with a 'Union' step that combines 'hb14_hb19' and 'special-health-boards_19022021'. This is followed by 'Join 1', 'Join 3', 'Join 5', and finally a 'Clean 1' step. The bottom pane is divided into 'Settings' and 'Union Results'. The 'Settings' pane shows 'Inputs' as 'hb14_hb19' and 'special-health-boards_19022021', and 'Resulting Fields' as '2 Mismatching fields from 4 resulting fields'. The 'Union Results' pane shows a table with columns: 'Table Names', 'Country', 'SHB', and 'SHBName'. The data rows show a list of hospital names and their corresponding SHB codes.

Table Names	Country	SHB	SHBName
hb14_hb19.xlsx/hb14_hb19	S92000003	S08000015	Healthcare Impr...
special-health-boards_1902...		S08000016	National Facility
		S08000017	NHS Ayrshire and
		S08000018	NHS Borders
		S08000019	NHS Dumfries and
		S08000020	NHS Education for
		S08000021	NHS Fife
		S08000022	NHS Forth Valley
		S08000023	NHS Grampian

Step Two

Tableau Prep Builder - CWT

File Edit Flow Server Help

Alerts (0)

Connections

- hb14_hb19.xlsx
- special-health-boar...
- current-hospital fla...
- cwt_62_day_stand...
- geography_codes_a...

Tables

- geography_codes_an...

Join 1: 5 fields, 27 rows

Settings

Applied Join Clauses

Country = Country

Join Type: right

Summary of Join Results

Join Clauses

Join 1

Country

Join Results

Table Names	Country	SHB
hb14_hb19.xlsx/hb14_hb19	S92000003	S08000015
special-health-boards_1902...		S08000016
		S08000017
		S08000018
		S08000019
		S08000020
		S08000021
		S08000022
		S08000023

Step Three

Tableau Prep Builder - CWT

File Edit Flow Server Help

Alerts (0)

Connections

- hb14_hb19.xlsx
- special-health-boar...
- current-hospital fla...
- cwt_62_day_stand...
- geography_codes_a...

Tables

- geography_codes_an...

Join 3: 10 fields, 1K rows

Settings

Applied Join Clauses

Join 1: SHB = HB

Join Type: left

Summary of Join Results

Join Clauses

Join 1

SHB

Join 3

HB

Join Results

Table Names	Country	SHB
hb14_hb19.xlsx/hb14_hb19	S92000003	S08000015
		S08000016
		S08000017
		S08000019
		S08000020
		S08000022
		S08000024
		S08000025
		S08000026

Step Four

Tableau Prep Builder - CWT

File Edit Flow Server Help

Alerts (0)

Connections

- hb14_hb19.xlsx
- special-health-boar...
- current-hospital fla...
- cwt_62_day_stand...
- geography_codes_a...

Tables

- geography_codes_an...

Join 3: 10 fields, 1K rows

Settings

Change Type

Rename Field

Filter: Range of Dates

Remove Field

Remove Field

Remove Field

Remove Field

Remove Field

Join

Filter

Join Clauses

Join 1

SHB

Join 3

HB

Join Results

Table Names	Country	SHB	SHBName
hb14_hb19.xlsx/hb14_hb19	S92000003	S08000015	
		S08000016	
		S08000017	
		S08000019	
		S08000020	
		S08000022	
		S08000024	
		S08000025	
		S08000026	
		S08000028	
		S08000029	
		S08000030	
		S08000031	
		S08000032	

Step Five

Tableau Prep Builder - CWT
File Edit Flow Server Help

Connections

- hb14_hb19.xlsx (Microsoft Excel)
- special-health-boar... (Microsoft Excel)
- current-hospital_fla... (Text File)
- cwt_62_day_stand... (Text File)
- geography_codes_a... (Text File)

Tables

- geography_codes_an...

Join 5: 11 fields, 22K rows

Applied Join Clauses

Join 3: HB = current-hospital_fla...

Join Type: Inner

Summary of Join Results

Join Clauses

Join 3: HB

Join Results

Postcode - Split 1: 170

SHB: 14

HEALTHBOARD

Step Six

Tableau Prep Builder - CWT
File Edit Flow Server Help

Connections

- hb14_hb19.xlsx (Microsoft Excel)
- special-health-boar... (Microsoft Excel)
- current-hospital_fla... (Text File)
- cwt_62_day_stand... (Text File)
- geography_codes_a... (Text File)

Tables

- geography_codes_an...

Clean 1: 8 fields, 22K rows

Changes (5)

Postcode: 170

HEALTHBOARD: 14

CountryName: 1

TIME: 40

CancerType: 1

Output

Postcode	HEALTHBOARD	CountryName	TIME	CancerType	NumberOfEligibleReferrals62DayStandard	NumberOfEligibleReferralsTreatedWithin62Days
KA27	NHS Ayrshire and Arran	Scotland	10/01/2021	Urological	3	1

Step Seven:

Tableau Prep Builder - CWT
File Edit Flow Server Help

Connections

- hb14_hb19.xlsx (Microsoft Excel)
- special-health-boar... (Microsoft Excel)
- current-hospital_fla... (Text File)
- cwt_62_day_stand... (Text File)
- geography_codes_a... (Text File)

Tables

- geography_codes_an...

Output: 8 fields

Save output to

File

Browse

Name

Output

Location

C:\Users\user\OneDrive\Documents\My Tableau Prep Repository\Datasources

Output type

Tableau Data Extract (.hyper)

Write Options

Select an option to create or update your output table.

Full refresh

Create table

Run Flow

Save to Output.hyper

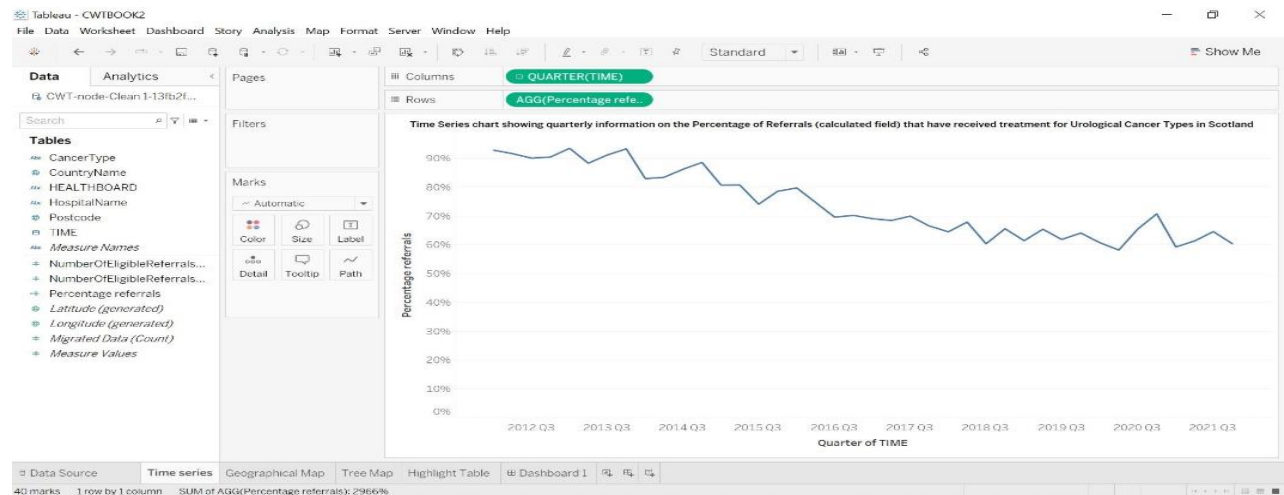
Postcode	HEALTHBOARD	CountryName	TIME	CancerType	NumberOfEligibleReferrals62DayStandard	NumberOfEligibleReferralsTreatedWithin62Days
KA27	NHS Ayrshire and Arran	Scotland	10/01/2021	Urological	3	1
KA27	NHS Ayrshire and Arran	Scotland	10/01/2021	Urological	42	42
KA27	NHS Ayrshire and Arran	Scotland	07/01/2021	Urological	4	3
KA27	NHS Ayrshire and Arran	Scotland	07/01/2021	Urological	28	28
KA27	NHS Ayrshire and Arran	Scotland	04/01/2021	Urological	2	1
KA27	NHS Ayrshire and Arran	Scotland	04/01/2021	Urological	48	48
KA27	NHS Ayrshire and Arran	Scotland	01/01/2021	Urological	3	3
KA27	NHS Ayrshire and Arran	Scotland	01/01/2021	Urological	30	30
KA27	NHS Ayrshire and Arran	Scotland	10/01/2021	Urological	3	1
KA27	NHS Ayrshire and Arran	Scotland	10/01/2021	Urological	42	42
KA27	NHS Ayrshire and Arran	Scotland	07/01/2021	Urological	4	3
KA27	NHS Ayrshire and Arran	Scotland	07/01/2021	Urological	28	28
KA27	NHS Ayrshire and Arran	Scotland	04/01/2021	Urological	2	1
KA27	NHS Ayrshire and Arran	Scotland	04/01/2021	Urological	48	48

Dashboard

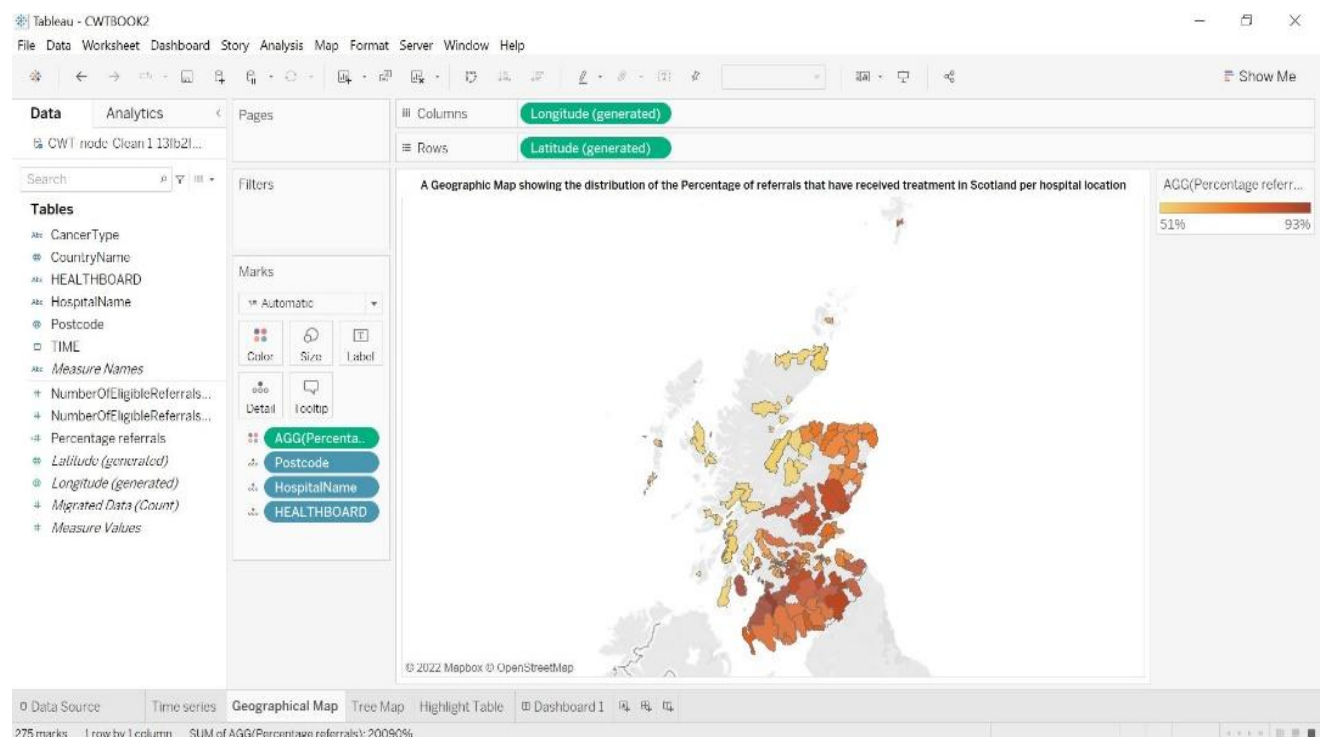
Access Link:

<https://public.tableau.com/app/profile/simi.oduba/viz/P2686827TABLEAUCWTDASHBOARD/P2686827TABLEAUDASHBOARD>

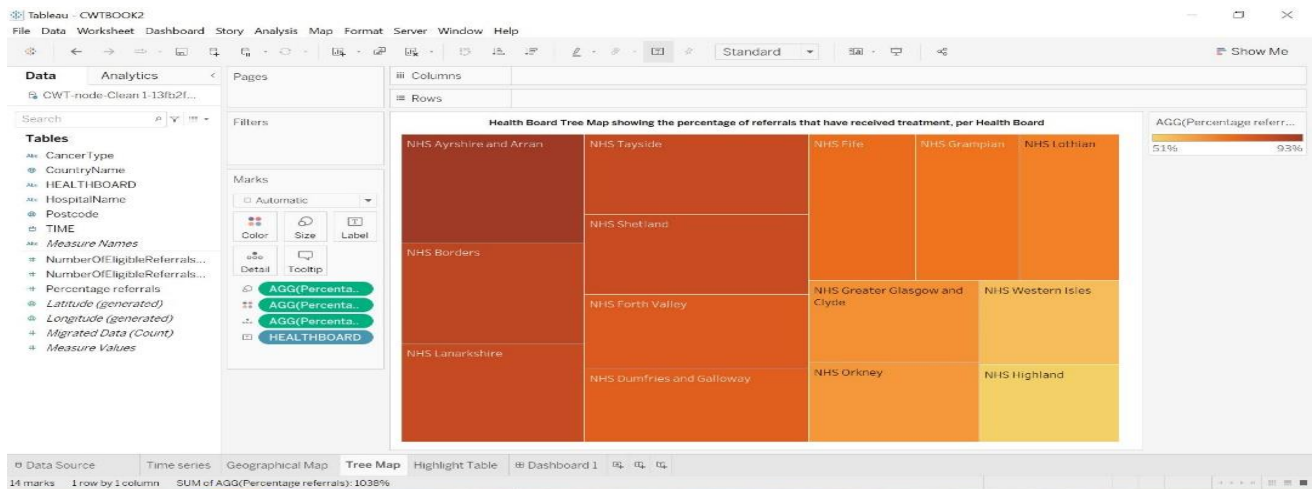
The statistics on urological cancer waiting times with a 62-day waiting standard is shown in the chart below for NHS Scotland health boards.



Scotland's percentage of referrals for various types of urological cancer is depicted in a time series chart..



Geographical map displaying the proportion of referrals treated at each hospital location in Scotland.



Tree Map for the Health Board displaying the percentage of treated referrals for each health Board.

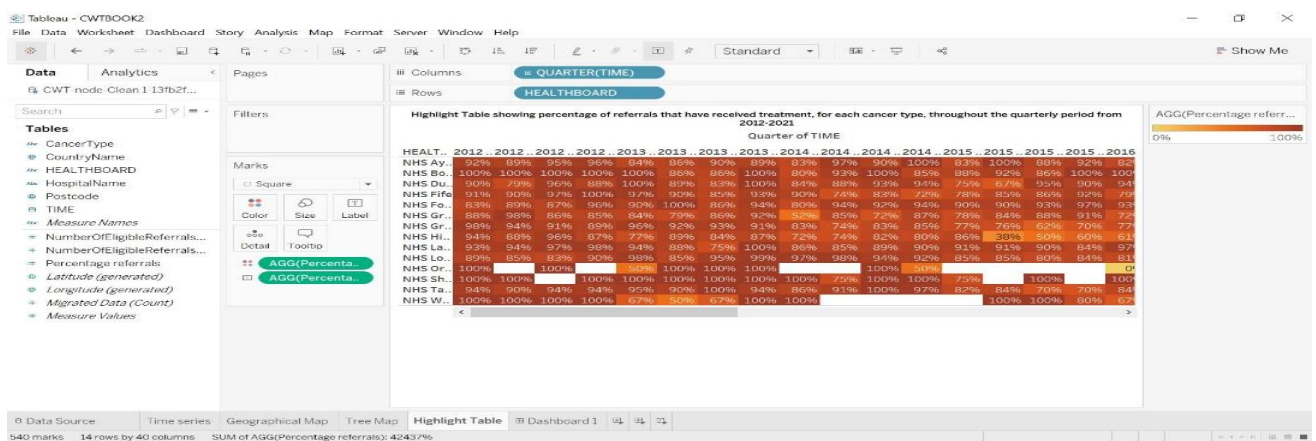
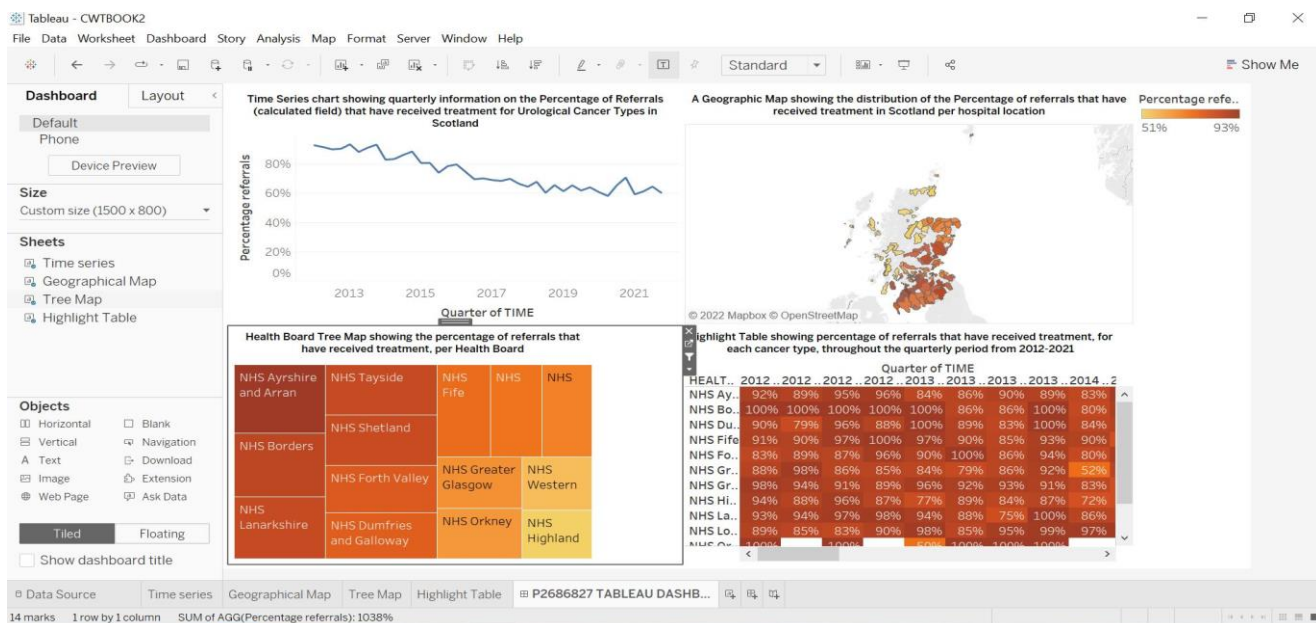


Table highlighting the proportion of referrals treated for each kind of cancer per quarter; 2012-2021.



Dashboard of the overall data

Explanation of Dashboard's output of NHS Scotland

The dashboard displays a time series chart, a geographic graph, a tree map of the health boards, and a table of the percentage of referrals treated for cancer in NHS Scotland. The time series graph that was found showed that the percentage of referrals treated among all health boards of the NHS Scotland experienced a negative trend series pattern from a high referral rate of 93% in 2012Q1 and only experienced an increase of 71% between Q3 and Q4 of 2020 the peak of the Covid period after which it reduced again to the range of 60% by the last quarter of 2021.

The geographical map indicated that mostly the Southern regions and the Eastern regions were the areas where a high number of referrals occurred with the Western and Northern regions having the lowest records of referrals. All the regions had referrals above average with the lowest referral rate being 51%.

The health board treemap shows with the help of the color gradient the deepest red color with health board NHS Ayrshire and Arran with 93% has the highest number of referrals out of all the health boards with NHS Highland having the lowest referrals of 51% patients. This can help the management know which health boards to focus on and redistribute their resources accordingly.

The percentage of referrals that have received treatment is displayed in the highlight Table. throughout the quarterly period from 2012-2021. It shows that show 3 health boards had a 0% referral rate for some quarters such as NHS Orkney, NHS Western Isles, and NHS Shetland while some health boards had 100% referral rates at certain quarters also including some who had 0% had at a point i.e NHS Western Isles, NHS Shetland, NHS Ayrshire and Arran, NHS Orkney. Some health boards also had missing values for certain quarters majorly NHS Orkney, NHS Western Isles, and NHS Shetland had few missing values.

This denotes that there has been a low improvement in the number of treated patients in Scotland in the time series chart of the percentage of referrals treated. It implies that there is a need for developing a significant improvement in increasing the percentage rate of referral treatment in the hospital through increasing the facilities and funding package for the NHS in Scotland. Through this analysis, the NHS Scotland has an overview of where the next phase of a strategy to be targeted i.e providing facilities or funding that can increase the percentage of referred treatment in Scotland overall.

PART 3: ETHICAL ISSUES IN BUSINESS INTELLIGENCE

Legal

Governmental bodies or policies exist to guide the treatment of patients with cancer in NHS Scotland. Who is held liable for dates prescribed to patients who end up not getting treated or who is in charge of the consequences of the actions or inactions of the health system, is it the management or the NHS? Great care must be made towards the actions and systems implemented by the NHS so as to avoid suits or legal issues arising in regard to patient's treatments and welfare.

Ethics

Ethics is a critical aspect of business intelligence due to the involvement of data in decision-making in any situation (Kemparaj & Kadalur, 2018). For example, in the case of NHS, the body expects to secure the data on the treatment of referred cancer patients to enable the process effectively because there will be cases of disagreement in the output. There should be informed consent of data use to patients and ensure the integrity of the NHS Scotland as a whole. Sharing of data with other agencies should be questioned and monitored so patient's records are not exploited.

Privacy issues

In this aspect, the Privacy of the data should be given priority through standard confidential processes. For instance, the analysis for NHS Scotland on the percentage of referred treated above reveals a negative trend that needs to be private to prevent escalating among opposition political parties or citizens. It should be of note that the NHS Scotland's data shouldn't be shared with 3rd parties and abused, and the privacy of information shouldn't be entirely generalized as patient's information vary from individual to individual.

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